

A Direct Proof of the Pipeline Form of the Generalized Brown Projection

Notes for the mccalum formalization project

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Abstract

These notes give a self-contained, direct proof of the *pipeline form* of the generalized Brown reduced projection: if P is a nonzero element of the elimination ideal $\mathcal{E}(f) = \langle f, f_z \rangle \cap R$, and both P and $\text{lc } f$ are order-invariant on a connected analytic submanifold S (with f vanishing identically nowhere on S), then f is degree-invariant on S . This is Corollary 5.3 of the companion notes *Generalizing Brown's Reduced McCallum Projection to Elimination Ideals*, where it is deduced from a general theorem (proved through a bivariate homogenization layer) together with an explicit Bézout “certificate transfer” lemma. Here we bypass both. The only role the second elimination ideal $\mathcal{E}(\text{rev } f)$ plays in the argument is to supply a single *membership* — $\text{lc}(f)^M P \in \mathcal{E}(\text{rev } f)$ — and membership, as opposed to an explicit certificate, has a three-line proof: reflect through the Laurent automorphism $z \mapsto 1/z$, then clear the spurious factor z^M by a one-line congruence rather than a binomial expansion. Inlining this into Brown’s pointwise reversal argument removes the homogenization layer entirely, leaving a proof that rests only on the already-formalized generalized lifting theorem and on elementary, cofactor-free polynomial algebra.

1 Setting and notation

Throughout, $R := \mathbb{R}[x_1, \dots, x_k] = \mathbb{R}[\bar{x}]$ and $f \in R[z]$ is a polynomial of positive degree $n := \deg_z f$, written $f = \sum_{i=0}^n a_i z^i$ with $a_i \in R$ and $a_n \neq 0$. We write $f_z := \partial f / \partial z$ and $\text{lc } f := a_n$ for the leading coefficient. For $q \in \mathbb{R}^k$ we write $f(q, \cdot) \in \mathbb{R}[z]$ for the specialization. The central object is the *elimination ideal*

$$\mathcal{E}(f) := \langle f, f_z \rangle \cap R,$$

the ideal of R consisting of polynomials in $\bar{x}$ alone expressible as $Af + Bf_z$ with $A, B \in R[z]$.

For a nonzero $g \in R$ and $p \in \mathbb{R}^k$, $\text{ord}_p g$ denotes the order of vanishing of g at p (the least total degree of a nonvanishing partial derivative). g is *order-invariant* on $S \subseteq \mathbb{R}^k$ if $p \mapsto \text{ord}_p g$ is constant on S , and *sign-invariant* on S if $p \mapsto \text{sign } g(p) \in \{-1, 0, +1\}$ is constant on S . Analytic submanifolds of $\mathbb{R}^k$ and analytic delineability are as in McCallum [2]; we emphasize that the formalized notion of delineability (**AnalyticDelineable**) carries *constant multiplicities*: the real roots of $f(q, \cdot)$ over S are given by finitely many analytic sections $\theta_1 < \dots < \theta_s$ on S , the $\theta_j(q)$ exhaust the real roots of $f(q, \cdot)$ for every $q \in S$, and the multiplicity of $\theta_j(q)$ as a root of $f(q, \cdot)$ is a constant $m_j \geq 1$.

The single analytic input is the following theorem, already proved in the repository (Lean: `lifting_theorem_generalized`, in `Mccalum/Generalized/Projection.lean`). It generalizes McCallum’s Theorem 2 [2] by replacing the discriminant with an arbitrary nonzero element of $\mathcal{E}(f)$.

Theorem 1.1 (Generalized lifting theorem, Theorem 3.2.1’; proved in Lean). *Let S be a connected analytic submanifold of $\mathbb{R}^k$, and let $f \in R[z]$ be degree-invariant on S with $f(q, \cdot) \not\equiv 0$*

for every $q \in S$. Suppose there is $P \in R$, $P \neq 0$, with $P \in \mathcal{E}(f)$, such that P is order-invariant on S . Then f is analytically delineable on S and is order-invariant in each f -section over S .

Everything below uses Theorem 1.1 exactly once, applied not to f itself but to an auxiliary polynomial built from f .

2 Brown's theorem and its naive generalization

McCallum's projection of f contains *all* coefficients $a_n, \dots, a_0$ (to secure degree-invariance on cells), together with the discriminant and pairwise resultants. Brown's contribution [1, Theorem 3.1] is that the non-leading coefficients are redundant:

Theorem 2.1 (Brown, 2001). *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z , with discriminant $D(\bar{x}) \neq 0$. Let S be a connected analytic submanifold of $\mathbb{R}^k$ on which D is order-invariant and $\text{lc } f$ is sign-invariant, and such that f vanishes identically at no point of S . Then f is degree-invariant on S .*

Degree-invariance is exactly the hypothesis Theorem 1.1 needs; combining the two yields Brown's *reduced* projection: leading coefficients, discriminants, and resultants only. Given the repository's generalization of McCallum's theorem, the natural candidate statement is:

Conjecture 2.2 (Naive generalization — **false**). *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z . Let $P \in \mathcal{E}(f)$, $P \neq 0$. Let S be a connected analytic submanifold of $\mathbb{R}^k$ on which P is order-invariant and $\text{lc } f$ is sign-invariant, and such that f vanishes identically at no point of S . Then f is degree-invariant on S .*

The hypothesis already implies $D \neq 0$: if f had a repeated factor over $\mathbb{R}(\bar{x})$, then $\gcd(f, f_z)$ would have positive z -degree and divide every element of $\langle f, f_z \rangle$ in $\mathbb{R}(\bar{x})[z]$, forcing $\mathcal{E}(f) = 0$. So Conjecture 2.2 genuinely only relaxes the *order-invariance* hypothesis from D to a general $P \in \mathcal{E}(f)$, in exact analogy with the proved Theorem 1.1.

3 A counterexample

Proposition 3.1. *Conjecture 2.2 is false. Let $k = 2$, $\bar{x} = (x, y)$, and*

$$f = -xy^2z^2 + yz + 1, \quad n = 2, \quad \text{lc } f = -xy^2,$$

let $S = \{(0, t) : t \in \mathbb{R}\}$ be the y -axis, and let $P = 1 + 4x$. Then all hypotheses of Conjecture 2.2 hold, but f is not degree-invariant on S .

Proof. First, $P \in \mathcal{E}(f)$, with the explicit certificate

$$1 + 4x = (1 - 2xyz + 4x) \cdot f + (-z(1 + 2x - xyz)) \cdot f_z, \quad (1)$$

verified by direct expansion. Clearly $P \neq 0$. S is a connected analytic submanifold of $\mathbb{R}^2$. On S we have $P(0, t) = 1$, so P is nowhere zero on S and hence order-invariant (constant order 0). The leading coefficient $-xy^2$ vanishes identically on S , hence is sign-invariant (constant sign 0). Finally $f(0, t, z) = tz + 1$ is never the zero polynomial. Yet $\deg_z(tz + 1)$ equals 1 for $t \neq 0$ and 0 for $t = 0$: the degree jumps at the origin. $\square$

Remark 3.2 (Why the corrected hypothesis excludes it). The degree drop at the origin is a collision of roots *at infinity* ($tz + 1$ has its root $-1/t \rightarrow \infty$ as $t \rightarrow 0$), invisible to $\langle f, f_z \rangle \cap R$; indeed $V_{\mathbb{C}}(\mathcal{E}(f)) = \{x = -1/4\}$, disjoint from the complexification of S , which is why a unit such as $1 + 4x$ can lie in $\mathcal{E}(f)$ along S . The theorem of this note (Theorem 5.3) excludes

Proposition 3.1 not through the witness P — which *is* order-invariant on S — but through the additional hypothesis that $\text{lc } f$ be order-invariant. On S the leading coefficient $-xy^2$ has

$$\text{ord}_{(0,t)}(-xy^2) = 1 \quad (t \neq 0), \quad \text{ord}_{(0,0)}(-xy^2) = 3,$$

since $-xy^2 = -t^2x - 2tx(y-t) - x(y-t)^2$ near $(0, t)$ has lowest-degree part $-t^2x$ (degree 1) when $t \neq 0$, but is the pure degree-3 monomial at the origin. So $\text{lc } f$ is *not* order-invariant on S , and Theorem 5.3 correctly refuses the configuration.

4 Reversal and the witness transfer

The proof in Section 5 is Brown's pointwise reversal argument. Its one nontrivial ingredient is a witness in the elimination ideal of the reversed auxiliary polynomial. This section produces it from $P \in \mathcal{E}(f)$ by elementary means: a reflection step and a clearing step, neither of which requires exhibiting Bézout cofactors.

Definition 4.1. For $g = \sum_{i=0}^m c_i z^i \in R[z]$ with $\deg g \leq m$, the *degree- m reverse* is $\text{rev}_m g := \sum_{i=0}^m c_i z^{m-i} = z^m g(1/z)$ (Mathlib: `Polynomial.reflect m g`; for $m = \deg g$ this is `Polynomial.reverse`). For f as in Section 1 we write $\text{rev } f := \text{rev}_n f = a_0 z^n + \cdots + a_n$; note its constant coefficient is $a_n = \text{lc } f$.

We will use the Euler-type identity relating $\text{rev}_{n-1}(g_z)$ to the derivative of $\text{rev}_n g$ (reversal and d/dz do not commute).

Lemma 4.2 (Euler). *For $g \in R[z]$ with $\deg g \leq n$,*

$$\text{rev}_{n-1}(g_z) = n \cdot \text{rev}_n g - z(\text{rev}_n g)'.$$

Proof. Write $g = \sum_{i=0}^n c_i z^i$. Both sides equal $\sum_{i=0}^n i c_i z^{n-i}$: the left side because $g_z = \sum_i i c_i z^{i-1}$ reverses at degree $n-1$ to $\sum_i i c_i z^{(n-1)-(i-1)}$; the right side because $(\text{rev}_n g)' = \sum_i (n-i) c_i z^{n-i-1}$, so $n \text{rev}_n g - z(\text{rev}_n g)' = \sum_i (n - (n-i)) c_i z^{n-i}$. $\square$

4.1 Reflection

Lemma 4.3 (Reflection). *Let $g \in R[z]$ with $\deg g \leq n$ and put $\rho := \text{rev}_n g$. If $P \in \langle g, g_z \rangle$ (the ideal in $R[z]$), then there is an integer $M \geq 0$ with*

$$z^M P \in \langle \rho, \rho' \rangle \quad (\text{ideal in } R[z]).$$

Proof. Work in the Laurent ring $R[z^{\pm 1}]$, in which z is a unit, and let $\iota : R[z^{\pm 1}] \rightarrow R[z^{\pm 1}]$ be the R -algebra automorphism $z \mapsto z^{-1}$ (an involution fixing R pointwise). From $\rho = z^n g(1/z)$ we get $\iota(g) = g(1/z) = z^{-n} \rho$, and by Lemma 4.2, $\text{rev}_{n-1}(g_z) = z^{n-1} g_z(1/z) = z^{n-1} \iota(g_z)$ equals $n\rho - z\rho'$, so $\iota(g_z) = z^{1-n}(n\rho - z\rho')$. Since z is a unit,

$$\iota(\langle g, g_z \rangle) = \langle \iota(g), \iota(g_z) \rangle = \langle \rho, n\rho - z\rho' \rangle = \langle \rho, \rho' \rangle \quad \text{in } R[z^{\pm 1}].$$

Now $P \in \langle g, g_z \rangle$ in $R[z]$, hence in $R[z^{\pm 1}]$. Applying ι and using $\iota(P) = P$ (as $P \in R$),

$$P = \iota(P) \in \iota(\langle g, g_z \rangle) = \langle \rho, \rho' \rangle \quad \text{in } R[z^{\pm 1}].$$

Write $P = \alpha\rho + \beta\rho'$ with $\alpha, \beta \in R[z^{\pm 1}]$, and let M be larger than the largest negative power of z occurring in α or β . Then $z^M \alpha, z^M \beta \in R[z]$, so $z^M P = (z^M \alpha)\rho + (z^M \beta)\rho' \in \langle \rho, \rho' \rangle$ in $R[z]$. $\square$

4.2 Clearing the spurious power

Lemma 4.4 (Clearing). *Let $\rho \in R[z]$ with constant coefficient $c := \rho(0) \in R$, and write $\rho = zh + c$ (so $h = \text{div}_X \rho$). If $z^M P \in \langle \rho, \rho' \rangle$ in $R[z]$, then*

$$c^M P \in \langle \rho, \rho' \rangle \quad (\text{ideal in } R[z]).$$

Proof. Work modulo $\langle \rho \rangle$ in $R[z]$. From $\rho = zh + c$ we get $zh \equiv -c$, hence $z^M h^M \equiv (-c)^M = (-1)^M c^M \pmod{\langle \rho \rangle}$. Multiplying the hypothesis $z^M P \in \langle \rho, \rho' \rangle$ by $h^M \in R[z]$ gives $z^M h^M P \in \langle \rho, \rho' \rangle$, and since $z^M h^M P \equiv (-1)^M c^M P \pmod{\langle \rho \rangle}$, also $(-1)^M c^M P \in \langle \rho, \rho' \rangle$. Multiply by $(-1)^M$. $\square$

Conceptually, Lemma 4.4 says: inverting the constant coefficient c makes z a unit modulo ρ (its inverse is $-h/c$), so the z^M produced by reflection can be divided away, at the cost of a power of c . The factor is unavoidable in general (see Remark 6.2).

4.3 The transfer proposition

Proposition 4.5 (Witness transfer). *Let $g \in R[z]$ have degree exactly $n \geq 1$ in z , leading coefficient $\text{lc } g$, and put $\rho := \text{rev}_n g$ (so $\rho(0) = \text{lc } g$). If $P \in \mathcal{E}(g) = \langle g, g_z \rangle \cap R$, then there is $M \geq 0$ with*

$$\text{lc}(g)^M P \in \mathcal{E}(\rho) = \langle \rho, \rho' \rangle \cap R.$$

Proof. By Lemma 4.3, $z^M P \in \langle \rho, \rho' \rangle$ for some M . The constant coefficient of $\rho = \text{rev}_n g$ is the coefficient of z^n in g , namely $\text{lc } g$. By Lemma 4.4 (with $c = \text{lc } g$), $\text{lc}(g)^M P \in \langle \rho, \rho' \rangle$; and $\text{lc}(g)^M P \in R$ since $\text{lc } g$ and P both lie in R . $\square$

Remark 4.6 (Verification). Both steps were checked symbolically on the data of Proposition 3.1 ($g = f$, $n = 2$, and the certificate (1) with $\deg_z A = 1$, $\deg_z B = 2$, giving $M = 3$): the reflection identity $z^3 P = U\rho + V\rho'$ holds with zero residual, and $\text{lc}(f)^3 P = -4x^4 y^6 - x^3 y^6 \in \langle \text{rev } f, (\text{rev } f)' \rangle$ with the cofactors produced by the congruence in Lemma 4.4, again with zero residual.

5 The direct proof

Brown's argument is local on S and passes through a point-dependent shift. We need his local connectivity-refinement lemma, an easy consequence of the straightening chart already formalized (`IsAnalyticSubmanifold.straightening_chart`).

Lemma 5.1 (Refinement). *Let S be a connected analytic submanifold of $\mathbb{R}^k$, $p \in S$, and N_0 an open neighborhood of p . There is an open N with $p \in N \subseteq N_0$ such that $S \cap N$ is a connected analytic submanifold of $\mathbb{R}^k$.*

Proof sketch. Take a straightening chart Φ at p carrying S locally onto a piece of $\mathbb{R}^s \times \{0\}$, shrink its source into N_0 , and pull back a small open ball B around $\Phi(p)$; then $\Phi(S \cap N) = B \cap (\mathbb{R}^s \times \{0\})$ is convex, hence connected, and the submanifold property is local. (The same argument appears inline in `Mccalum/Generalized/Delineable.lean`.) $\square$

We also record the shift transport, which is immediate from the chain rule.

Lemma 5.2 (Shift). *Let $\gamma \in \mathbb{R}$ and $\bar{f} := f(\bar{x}, z + \gamma)$. Then $\deg_z \bar{f} = n$, $\text{lc } \bar{f} = \text{lc } f$, and $\mathcal{E}(f) \subseteq \mathcal{E}(\bar{f})$; in particular $P \in \mathcal{E}(f) \Rightarrow P \in \mathcal{E}(\bar{f})$.*

Proof. The R -algebra automorphism $\tau : R[z] \rightarrow R[z]$, $z \mapsto z + \gamma$, fixes R pointwise, has $\tau(f) = \bar{f}$, and commutes with d/dz , so $\tau(f_z) = (\bar{f})_z$. It preserves z -degree and the leading coefficient. Applying τ to a representation $P = Af + Bf_z$ and using $\tau(P) = P$ gives $P = \tau(A)\bar{f} + \tau(B)(\bar{f})_z \in \langle \bar{f}, (\bar{f})_z \rangle$. $\square$

Theorem 5.3 (Generalized Brown, pipeline form). *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z . Let $P \in \mathcal{E}(f)$, $P \neq 0$, and let S be a connected analytic submanifold of $\mathbb{R}^k$ on which both P and $\text{lc } f$ are order-invariant, with $f(q, \cdot) \not\equiv 0$ for every $q \in S$. Then f is degree-invariant on S .*

Proof. Reduction. On the connected set S , order-invariance of $a_n = \text{lc } f$ forces a dichotomy: if its constant order is 0 then a_n vanishes nowhere on S , and then $\deg_z f(q, \cdot) = n$ for all $q \in S$ and we are done; if the constant order is ≥ 1 then $a_n \equiv 0$ on S . Assume the latter, so $\deg_z f(q, \cdot) \leq n - 1$ on S . Since $q \mapsto \deg_z f(q, \cdot)$ is integer-valued on the connected S , it suffices to show it is locally constant on S , i.e. constant within S near each $p \in S$.

Pointwise setup. Fix $p \in S$. As $f(p, \cdot) \not\equiv 0$ has finitely many roots, choose $\gamma \in \mathbb{R}$ with $f(p, \gamma) \neq 0$; by continuity of $q \mapsto f(q, \gamma)$ there is an open $N_0 \ni p$ with $f(q, \gamma) \neq 0$ for all $q \in N_0$. Put $\bar{f} := f(\bar{x}, z + \gamma)$ and $f^* := \text{rev}_n \bar{f}$, and write $\bar{f} = \sum_{i=0}^n \bar{a}_i z^i$, so that $\bar{a}_n = \text{lc } \bar{f} = a_n$, $\bar{a}_0 = \bar{f}(\bar{x}, 0) = f(\bar{x}, \gamma)$, and $f^* = \sum_{i=0}^n \bar{a}_i z^{n-i}$. We record three facts.

- (1) f^* is degree-invariant of degree n on N_0 : the coefficient of z^n in f^* is $\bar{a}_0 = f(\bar{x}, \gamma)$, nonzero on N_0 . In particular $f^*(q, \cdot) \not\equiv 0$ on N_0 .
- (2) *Multiplicity of 0 records the degree drop:* for $q \in N_0$,

$$\text{mult}_0 f^*(q, \cdot) = n - \max\{i : \bar{a}_i(q) \neq 0\} = n - \deg_z \bar{f}(q, \cdot) = n - \deg_z f(q, \cdot),$$

the last step because shifting z preserves degree. The constant coefficient of f^* is $\bar{a}_n = a_n$; so for $q \in S$ (where $a_n \equiv 0$), 0 is a root of $f^*(q, \cdot)$, of multiplicity $n - \deg_z f(q, \cdot) \geq 1$.

- (3) *Witness:* by Lemma 5.2, $P \in \mathcal{E}(\bar{f})$; applying Proposition 4.5 to $g = \bar{f}$ (degree n , with $\text{rev}_n \bar{f} = f^*$ and $\text{lc } \bar{f} = \text{lc } f$) yields $M \geq 0$ with

$$Q := \text{lc}(f)^M P \in \mathcal{E}(f^*) = \langle f^*, (f^*)_z \rangle \cap R.$$

Moreover $Q \neq 0$ (as $\text{lc } f \neq 0$, $P \neq 0$, and R is a domain), and Q is order-invariant on S , because orders add over products: $\text{ord}_q Q = M \text{ord}_q(\text{lc } f) + \text{ord}_q P$ is constant on S by hypothesis.

Lifting. By Lemma 5.1, choose an open $N \subseteq N_0$ with $p \in N$ and $S \cap N$ a connected analytic submanifold. By (1)–(3), all hypotheses of Theorem 1.1 hold for f^* on $S \cap N$ (degree-invariant of degree n ; $f^*(q, \cdot) \not\equiv 0$; witness $Q \in \mathcal{E}(f^*)$ nonzero and order-invariant). Hence f^* is analytically delineable on $S \cap N$: there are analytic sections $\theta_1 < \dots < \theta_s$ on $S \cap N$ exhausting the real roots of $f^*(q, \cdot)$, with constant multiplicities $m_1, \dots, m_s \geq 1$.

Tracking the zero section. By (2), 0 is a real root of $f^*(p, \cdot)$, so $\theta_i(p) = 0$ for exactly one index i (the $\theta_j(p)$ are pairwise distinct). By continuity of the finitely many sections, shrink N to an open $N' \ni p$ with $\theta_j \neq 0$ on $S \cap N'$ for all $j \neq i$. Let $q \in S \cap N'$ be arbitrary: by (2), 0 is a root of $f^*(q, \cdot)$, so $0 = \theta_j(q)$ for some j , forcing $j = i$. Therefore

$$n - \deg_z f(q, \cdot) = \text{mult}_0 f^*(q, \cdot) = \text{mult}_{\theta_i(q)} f^*(q, \cdot) = m_i,$$

so $\deg_z f(q, \cdot) = n - m_i$ for every $q \in S \cap N'$. This is the required local constancy, completing the proof. $\square$

6 Discussion

Remark 6.1 (Recovering Brown on CAD cells). Theorem 5.3 subsumes Brown’s Theorem 2.1 on the cells of a McCallum-style CAD. There the discriminant D is a projection factor, hence $D \in \mathcal{E}(f)$ (the classical fact $\text{disc}_n \in \langle \cdot, \cdot' \rangle$, used in the repository to recover Theorem 3.2.3 from 3.2.3’) and D is order-invariant on every cell; and $\text{lc } f$ is also a projection factor, hence order-invariant on every cell. So taking $P = D$, the hypotheses of Theorem 5.3 hold on each cell, and we recover degree-invariance — which is precisely Brown’s reduced projection in use. More generally the witness $\tilde{P} := \text{lc}(f)^M P$ of Brown’s reduced projection is never computed by the algorithm: $V(\tilde{P}) = V(\text{lc } f) \cup V(P)$ contributes no irreducible factor not already present, so no new cells, degrees, or projection work arise.

Remark 6.2 (The factor is necessary). Some power of $\text{lc } f$ is unavoidable in Proposition 4.5: Proposition 3.1 exhibits $P \in \mathcal{E}(f)$ with $P \notin \mathcal{E}(\text{rev } f)$, so there is no general inclusion $\mathcal{E}(f) \subseteq \mathcal{E}(\text{rev } f)$. The factor forces the witness to vanish on the degree-drop locus $\{a_n = a_{n-1} = 0\}$ — exactly the information at infinity that $\mathcal{E}(f)$ alone lacks. It is also what couples the conclusion to the hypothesis “ $\text{lc } f$ order-invariant”: order-invariance of $\text{lc } f$ is what makes $\text{lc}(f)^M P$ order-invariant in step (3).

References

- [1] C. W. Brown, *Improved projection for cylindrical algebraic decomposition*, J. Symbolic Computation **32** (2001), 447–465.
- [2] S. McCallum, *An improved projection operator for cylindrical algebraic decomposition*, in: Caviness, Johnson (eds.), *Quantifier Elimination and Cylindrical Algebraic Decomposition*, Springer, 1998.